Adenocarcinoma of the Esophagus and GE Junction

1

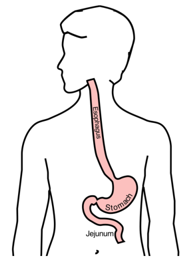
Anatomy

Food moves from the throat

→ esophagus

→ stomach

→ small bowel (jejunum)



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Cancer Staging

Staging refers to the tests to determine

- How large is the tumor?
- Has there been spread to lymph nodes?
- Has it spread to other parts of the body?

Treatment options depend upon the cancer stage

3

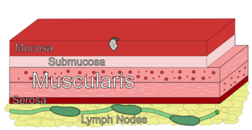
Cancer Staging

- **T** = Tumor - Depth of growth into the wall
- **N** = Nodes - Spread to the lymph nodes
- **M** = Metastasis - Spread to liver, lungs, or bone

4

Early Stage Cancers

Cancers start on the very inside layer called the mucosa



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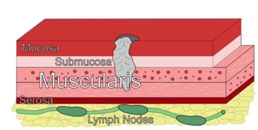
Locally-advanced Cancers

Over time, cancers can grow into the muscular wall



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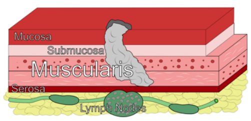
Lymph Nodes

In some cases, cancer cells can break off from the main tumor and spread to lymph nodes



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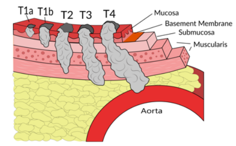
T Stage

Cancers are categorized based upon the thickness of the tumor, known as the T stage



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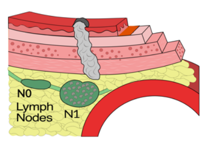
N Stage

Cancers are categorized by whether there is spread to the nodes.

- **N0** cancers have not spread to the nodes
- **N1** cancers have spread to the nodes.



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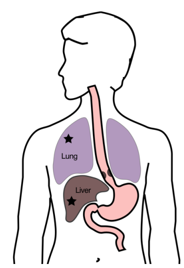
M Stage

Some cancers spread to other parts of the body

- **M0** cancers have not spread
- **M1** cancers have spread to lungs, liver, bone, or within the abdomen



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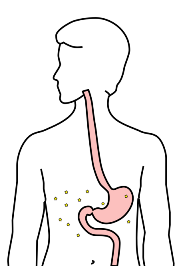
Carcinomatosis (M1)

Spread within the abdomen is **carcinomatosis**

Tumors can be very small (grain of rice)

Fluid can also build up: **ascites**



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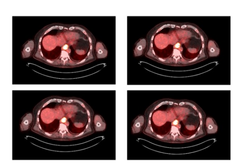
PET scan

Similar to CT scan

Tracer shows 'hot spots'

- Cancer
- Inflammation or infection
- Normal organs (heart, kidneys)



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PET scan Preparation

- Nothing to eat or drink for 4 hours prior to exam
- High-protein low-carbohydrate diet the day prior
- No strenous activity day prior to study

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PET scan Prepration - Diabetic Patients

PET scan uses a modified sugar as a tracer ⇒ you will need to be careful about blood sugar and insulin before the study

- No insulin within 4 hours of the exam
- OK to take diabetes pills the day of the exam
- PET scans are generally best in the morning
- Plan to eat a meal after the PET scan, with half the usual morning dose of insulin

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PET scan Prepration - Diabetic Patients

- If PET is scheduled after 10AM:
  - low-carbohydrate breakfast at least 4 hours before
  - half the usual regular (short-acting) insulin
- No long-acting insulin after midnight.
- Insulin pump: turn it off 4 hours before the study (or turn it down to basal/night-time rate)

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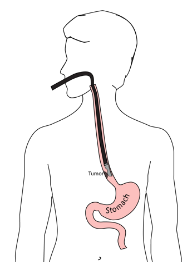
Endoscopic Ultrasound

- Similar to upper endoscopy (EGD)
- Ultrasound in scope
- Evaluates T stage



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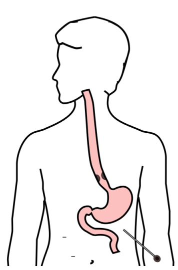
Laparoscopy

- Some stomach cancers can spread inside the abdomen
- Areas of spread can be very small (grain of rice)
- Laparoscopy can detect spread inside the abdomen



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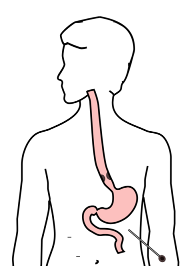
Laparoscopy

- General anesthetic
- 1/4” incisions
- Telescope examines the abdomen
- Biopsies performed
- Fluid sampled



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Treatment Plan

Superficial (T1)
- Endoscopic Therapy

Localized (T1b/T2)
- Surgery (esophagectomy)

Locally-advanced (T3M0)
- Chemo$\pm$Radiation $\rightarrow$Surgery

Metastatic (M1)
- Chemotherapy

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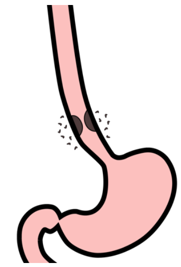
Locally-advanced cancers

Patients with locally-advanced esophageal cancer often have localized spread of cancer cells in the surrounding area



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Locally-advanced cancers

If locally-advanced cancers are treated with surgery alone…

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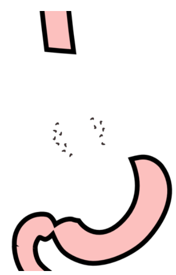
Locally-advanced cancers

If locally-advanced cancers are treated with surgery alone…

There is a risk that cancer cells can be left behind



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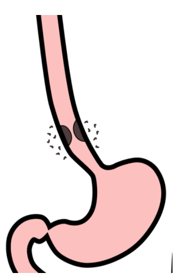
Preoperative Therapy

It is helpful to start with therapy *before* surgery that will shrink the cancer.



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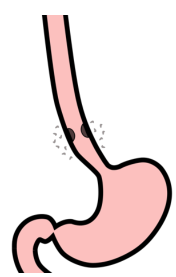

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Preoperative Therapy

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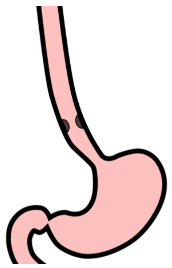

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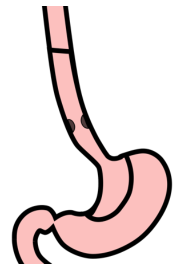
Surgery after Preoperative Therapy

When surgery is then performed…



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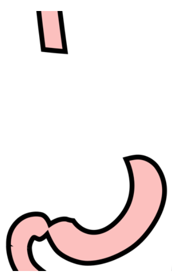
Preoperative Therapy

When surgery is then performed…

The risk of cancer recurrence is minimized



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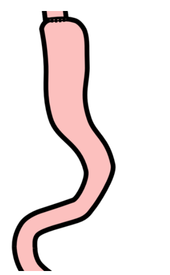
Preoperative Therapy

When surgery is then performed…

The risk of cancer recurrence is minimized



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Chemotherapy + Radiation CROSS Trial

363 patients with esophageal cancer studied

Patients were treated in two groups:

Surgery Alone

vs

Chemo + Radiation → Surgery

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Chemotherapy + Radiation CROSS Trial

363 patients with esophageal cancer studied

Chemotherapy + radiation together over 6 wks

Surgery Alone

vs

Chemo + Radiation→Surgery ⇒ Longer Survival

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Chemo + Radiation CROSS Trial

Typical schedule for chemotherapy + radiation:
- Chemotherapy once per week for six weeks
- Radiation five days per week for six weeks (28)
- PET scan (or CT) 4 weeks after the end of radiation
- Surgery 8 weeks after the end of radiation

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Chemo + Radiation - Side Effects

Kills cancer cells in the esophagus and lymph nodes
Can also irritate the lining of the esophagus.
Swallowing can be difficult the last 2 weeks.
Feeding tube may be needed for hydration/nutrition.

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Locally-advanced Adenocarcinoma

“Sandwich” chemotherapy before + after surgery:
Chemo (8 wks) → Surgery → Chemo (8 wks)
Two different drug combinations:
- FLOT (more effective)
- FOLFOX (better tolerated)

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“Sandwich” Chemotherapy Drugs

FLOT
- 5-FU
- Leucovorion
- Oxaliplatin
- Taxotere

FOLFOX
- 5-FU
- Leucovorin
- Oxaliplatin

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FLOT Treatment Plan

- FLOT Chemo every 2 weeks x 4 (=8 weeks total)
- Surgery (4-6 weeks later)
- FLOT Chemo every 2 weeks x 4 (=8 weeks total)

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Durvalumab Immunotherapy

Cancer cells can turn off immune cells using a protein PD-L1
Darvalumab turns on immune cells by counteracting PD-L1
Durvalumab strengthens the immmune system to fight cancer

36

Matterhorn Trial

The Matterhorn clinical trial compared two types of treatment:

FLOT → Surgery → FLOT

FLOT + Durvalumab → Surgery → FLOT + Durvalumab

Better survival with addition of durvalumab to FLOT

Treatment with durvalumab depends upon approval from insurance company

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FLOT Chemo ± Durvalumab

FLOT Chemotherapy

- FLOT (8 weeks)
- Surgery (4-6 weeks later)
- FLOT (8 weeks)

FLOT + Durvalumab

- FLOT + Durvalumab (8 wks)
- Surgery (4-6 weeks later)
- FLOT + Durvalumab (8 wks)
- Durvalumab monthly

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Tumor Biomarkers - Pathology Tests

Show whether other drugs may be helpful:

- HER-2 → Herceptin can be helpful
- PD-L1 → Immunotherapy can be helpful
- MMR → Immunotherapy can be very helpful

Biomarkers reported in a separate pathology report

Your medical oncologist will review these with you

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Adenocarcinoma Treatment Options

Chemo + Radiation

- Chemo + Radiation (6wk)
- Surgery

Chemotherapy

- Chemo (8wk)
- Surgery
- Chemo (8wk)

40

Adenocarcinoma Treatment Options

Chemo + Radiation

- Longer track record
- Better tolerated
- Port usually placed
- Eating worse → better
- May need feeding tube

Chemotherapy

- More effective
- More side effects
- Port always required
- Eating slowly improves
- Less likely to need feeding tube

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Chemotherapy Administration

Most chemotherapy is administered by vein.

Several options exist to administer chemotherapy:

- Intravenous catheter in peripheral veins
- Peripheral Intravenous Central Catheter (PICC)
- Central Venous port

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Intravenous Catheter in Peripheral Vein ("IV")

- IV catheter placed in vein of hand or arm
- Allows administration of chemo and fluids
- Placed for each dose
- Removed that day
- Not suitable for FLOT chemo

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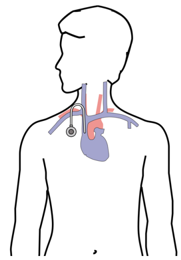
Central Venous Port

- Implantable device makes chemo easier
- May shower in 24 hrs
- No special care at home
- OK for FLOT chemo
- Allows for blood draws



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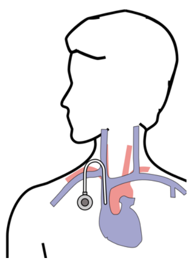
Central Venous Port

- Implanted under skin
- Neck incision (1/4")
- Incision below the collarbone
- Sutures dissolve
- "Superglue" on incisions



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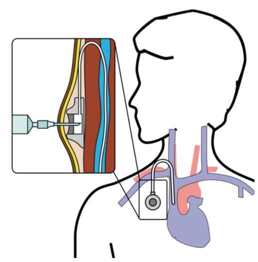
Central Venous Port

When it is time for chemotherapy, a needle is inserted through the skin into the port



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Restaging

CT or PET scan will be performed after preoperative therapy

- Surgery performed after restaging
- Timing depends upon recovery from therapy

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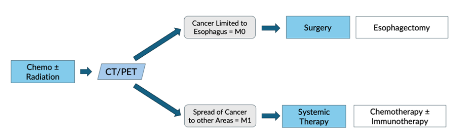
Restaging after Chemotherapy $\pm$ Radiation



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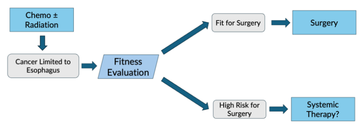
Fitness Evaluation prior to Surgery



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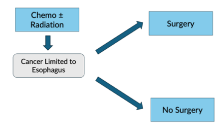
Surgery vs No Surgery



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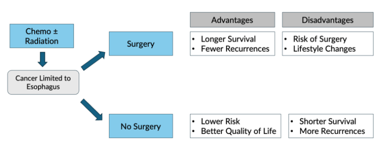
Surgery Advantages + Disadvantages



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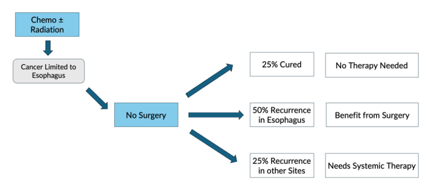
Outcomes after Chemotherapy $\pm$ Radiation



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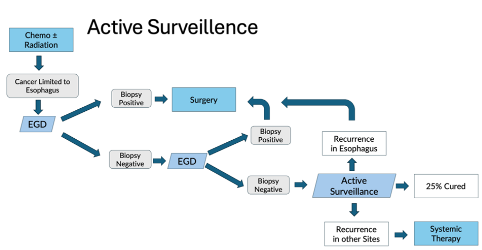

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Active Surveillance

Active Surveillance is appropriate when:

• CT/PET shows no signs of spread

• EGD biopsies are negative

Active Surveillance reduces the need for surgery by 50% *but* survival and disease control is probably better with immediate surgery.

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Primary Care Practitioner (PCP)

Critical to coordinate care between specialists.

We will update your PCP after each visit

PCP Referral Line (844) 235-6998

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My Atrium Patient Portal

- Critical to good communication with your care team
- Available for desktop or laptop or phone
- Sign up at my.atriumhealth.org

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Exercise

- Reduces risk of complications from treatment
- Goal is 30min/day of vigorous exercise 6 days/wk
  - Working hard enough that you can't converse
  - Start slowly and build up
  - Every day counts! (Aim for daily activity)

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Smoking Cessation

- Smoking makes cancer treatment more difficult
  - Increases risk of complications after surgery
- Options for help with smoking cessation:
  - NC Quit Line 1-800-QUIT-NOW (1-800-784-8669)
  - American Lung Assn www.freedomfromsmoking.org
  - Smoking Cessation Counseling (Metro Charlotte)

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Protein Needs

- Men: Average 75 grams/day
- Women: Average 60 grams/day

Protein Shakes provide protein with minimal sugar

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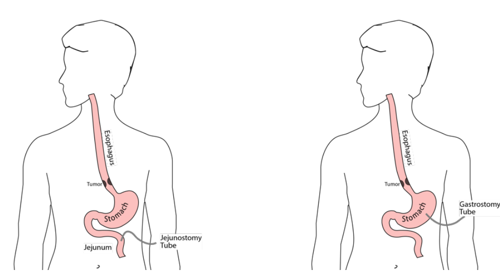
Feeding Tubes



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Gastrostomy Tube

Feeding Gastrostomy

- Feeding with a syringe several times per day.
- Tube can be hidden underneath clothing
- Tube does not interfere with eating by mouth
- Removed easily in the office when no longer needed

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Gastrostomy Tube Methods

PEG: Tube placed by endoscopy

Laparoscopic: Tube placed surgically by laparoscopy

Preferred method depends upon whether esophagectomy is planned

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Gastrostomy Tube

- Outpatient Placement (go home the same day)
- Central venous port can be placed at the same time (if needed)

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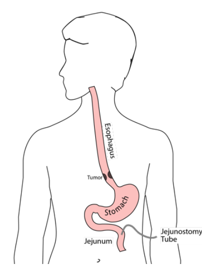
Jejunostomy Tube

- Nutrition to bypasses the esophagus and stomach
- Placed in small intestine
- Pump administers feedings slowly
- Feeding done at night



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Jejunostomy Typical Regimen

- Jejunostomy tube feeds for 16 hours (6pm-10am)
  - Men: 75mL/hour x 16 hours = 5 cartons
  - Women: 60mL/hour x 16 hours = 4 cartons
- Water 240ml (8oz) via syringe 4x/day

Hospital nurses will teach use of the feeding tube

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Jejunostomy Feeds with Diabetes

Jejunostomy feedings elevate blood sugars

- Insulin may be required along with feeds

Typical Pattern for tube feeds

- Feeds run via pump from 6pm to 10am
- Insulin at 6pm (70/30 insulin)
- Insulin at Midnight (70/30 insulin)
- No insulin if tube feedings are not run

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Jejunostomy Video

A video is available to help become familiar with the feeding jejunostomy

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Team Members - Physicians

Primary Care Provider

Gastroenterologist

Medical Oncologist (chemotherapy)

Radiation Oncologist (radiation)

Surgeons

- Jonathan Salo MD
- Jeffrey Hagen MD
- Michael Roach MD

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Team Members - Support Staff

Dietitian - Liz Koch

Nurses

- Matthew Carpenter RN
- Brandon Galloway LPN

Navigators

- Laura Swift
- Lorna Espinal

Care Coordinator - Karen Julian-Martinez

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